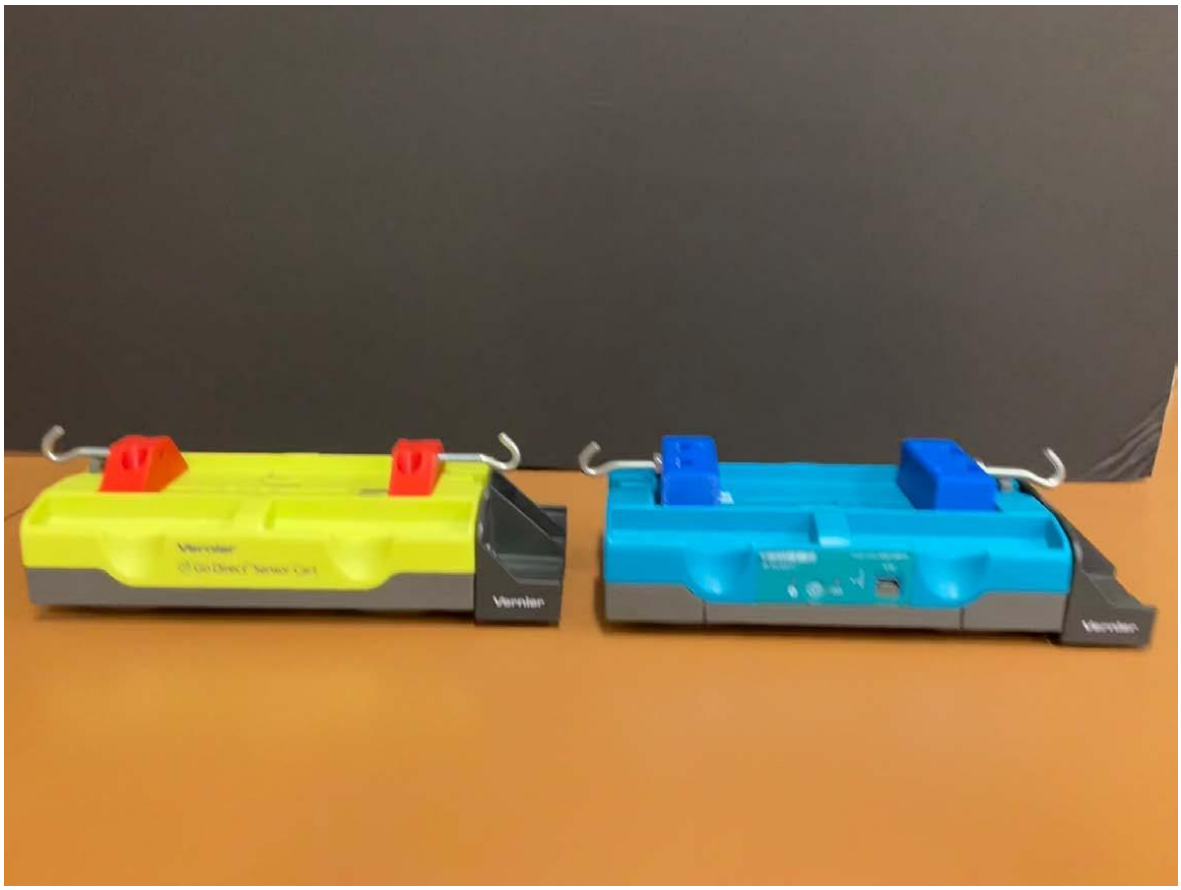
Free response of a spring-mass-damper system

MIE 402 - Mechanical Engineering Laboratory II
Fall 2026 | Prof. Ehsan Roohi



Course equipment: Vernier sensor carts. Use the cart assigned to your station; identify it by its serial number.

What you will determine

A suspension, machine mount or flexible instrument continues moving after a disturbance. This laboratory uses a spring-mounted cart to measure how fast that motion oscillates and how quickly it decays. You will measure the system properties, predict its natural frequency, and compare the prediction with recorded motion.

The workflow: identify and weigh hardware → check sensors → determine all four spring stiffnesses → zero the cart at equilibrium → collect seven configurations → analyze and show one complete case to the GTA before leaving.

Use **Vernier Graphical Analysis** to acquire data and fit the spring-calibration signal. Use **MATLAB** and **MIE402_Lab4_Analyze_Free_Response.m** to analyze exported cart CSV files. The cart measures position with a wheel encoder; the hanging-spring test measures force. These are two different measurements.

Canvas is the official source of submission dates. Keep the evidence for the combined Labs 4-5 report. The steps below preserve the supplied course procedure, including four individual springs and at least seven cart configurations.

Three operations that must not be confused

Operation	What it does	When to do it
Sensor zero / tare	Subtracts a constant offset. It does not determine the spring stiffness or change the sensor scale.	Force: unloaded, in its mounted orientation. Cart: resting at equilibrium.
Sensor calibration / verification	Checks the sensor reading against a known force or distance.	Before taking data; ask the GTA if the check fails.
Spring characterization	Determines each spring stiffness k [N/m] from its measured oscillation frequency and known hanging mass.	For each labeled spring A, B, C and D.

Prepare the station

1. Identify the cart, force sensor, track, four springs, hanging mass and hanger, hex-bar masses, magnetic brake hardware, balance and ruler. Label springs A-D without interfering with their motion.

2. Weigh and record all moving parts. For the hanging test, include the hanger and attachments that move with the mass. For cart tests, include the cart, added bars and every brake component that moves with it. Use kilograms in calculations: 100 g = 0.100 kg.

3. Secure the vertical support before hanging a mass. Keep the mass clear of the floor and bench. Do not use a hex bar as an improvised hanging mass; use the station-approved hanging weight and hook.

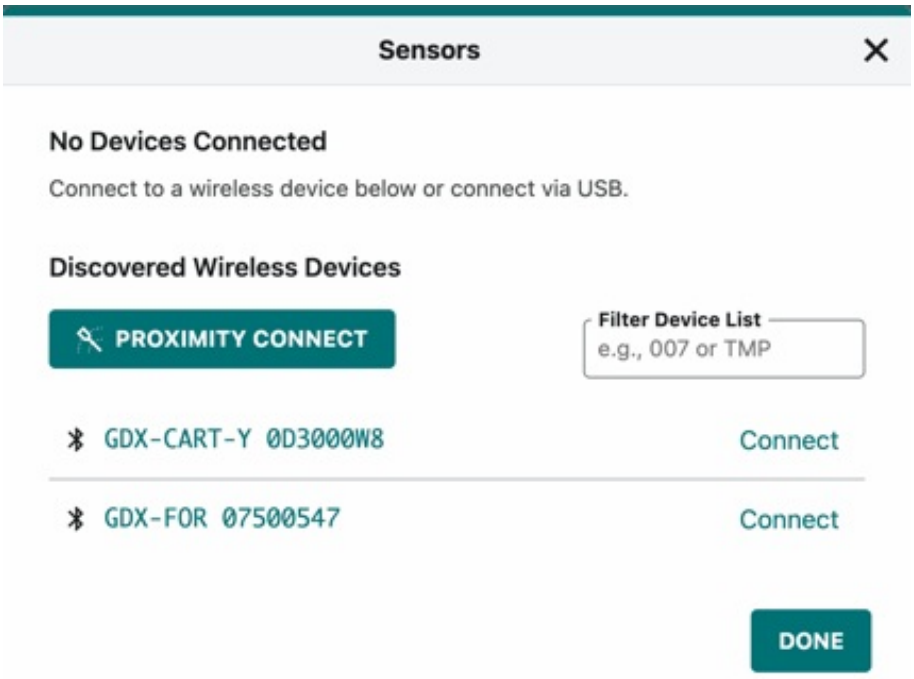
4. Level the track and check free rolling. Keep hands away from end stops and magnet gaps. Do not overstretch springs; the course setup uses approximately 15 cm spring lengths near cart equilibrium. Verify travel before every release.

Record before you start: group/section, date, cart and force-sensor serial numbers, spring IDs, measured masses, sensor units, sampling rate and filenames. Do not overwrite raw files.

What a successful setup looks like

The force sensor is rigidly supported, the spring hangs vertically and the mass oscillates without sideways swinging. The cart rolls along a level track, both springs remain taut throughout the planned motion, and the brake hardware clears the track.

Connect and check the force sensor



Graphical Analysis device list from the course notes. Select the force sensor serial number for spring tests, not a neighboring group's device or the cart position channel.

1. Power the **force sensor**. In Graphical Analysis choose **New Experiment → Sensor Data Collection**, or open the sensor connection icon. Match the displayed serial number to the sensor in your hand, connect, and select **Done**.

2. Display **Force [N] versus Time [s]**. Mount the sensor with its hook pointing downward, in the orientation used for the experiment. Let it settle with no hanging load. Open the force readout/sensor menu and choose **Zero**. Menu position varies with the installed app version.

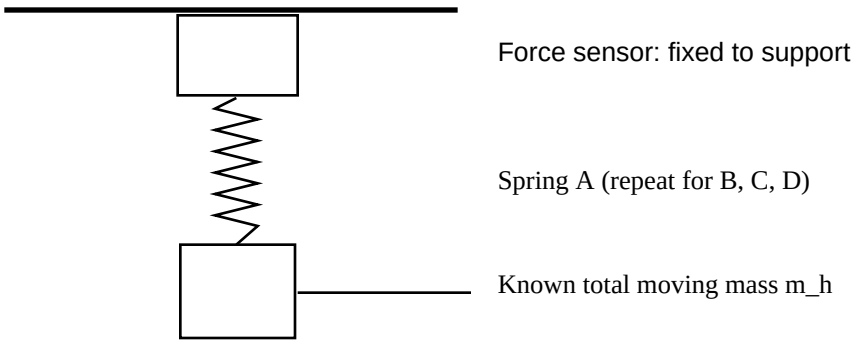
3. Check the scale with a measured, station-approved mass hung directly from the hook. For total hanging mass m_{check} [kg], expected force magnitude is $m_{check}g$ [N], with $g = 9.81$ m/s^2. Example only: 0.100 kg gives 0.981 N. Record expected and measured force, including sign.

4. Remove the check mass. The reading should return near zero. If the magnitude is inconsistent with the known weight beyond measurement uncertainty, check mass units, serial number, orientation and any applied load, then ask the GTA. Do not proceed by changing values until the reading looks right.

Zeroing is not a two-point calibration. Vernier force sensors have stored calibration. A failed check requires the GTA to identify the exact sensor model and follow its manual. Do not overwrite factory calibration or apply an arbitrary heavy calibration load. A reversed force sign is different from an incorrect scale.

If you zero with a spring already attached, its weight is included in the tare; note this. If you zero with the loaded mass attached, the oscillation frequency can still be fitted, but the mean force is no longer an absolute weight check. For a clear record, use the unloaded-zero sequence above.

Measure one spring: collect a usable force record



Small vertical motion only; keep the support stationary.

1. Hang spring A from the force-sensor hook. Attach a measured hanging mass securely to its free end. Record the **total moving hanging mass m_h**, including hanger and moving attachments. The force-sensor body is fixed and is not included in m_h.

2. Let the mass come to rest. Confirm that it clears the bench/floor and that the spring is not overextended. The load must produce a clear oscillation; ask the GTA to choose a safe mass rather than using an almost weightless hanger.

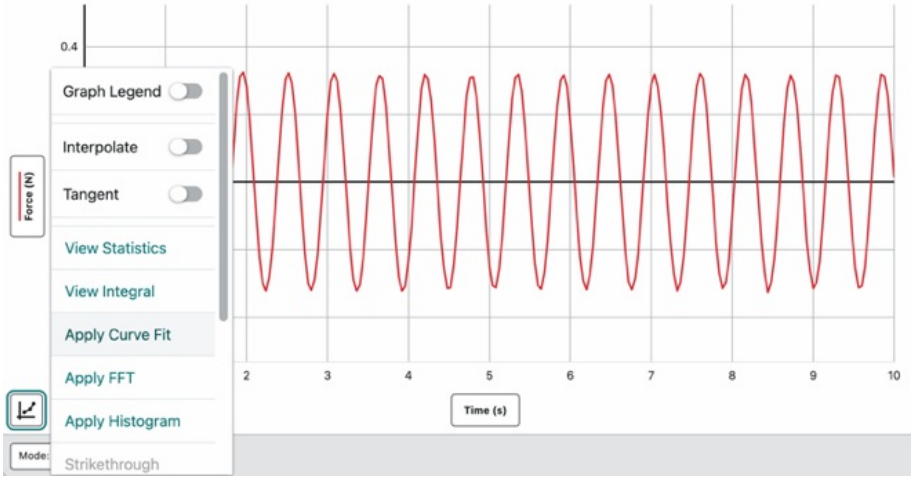
3. Open data-collection settings at the bottom left. Use **25 samples/s**, within the supplied course limit $f_s \leq 25$ Hz, and an initial **10 s** duration. Record the actual settings. Aim for roughly 15-20 samples/cycle and at least five clear cycles; if these conflict with the local sampling limit, consult the GTA about a safe heavier hanging mass or acquisition settings.

4. Start **Collect** before release. Pull the mass down slightly, then release gently without a push. Keep motion vertical. Stop after enough clean cycles; 5-10 s is the course starting range, but extend the duration if the frequency is low.

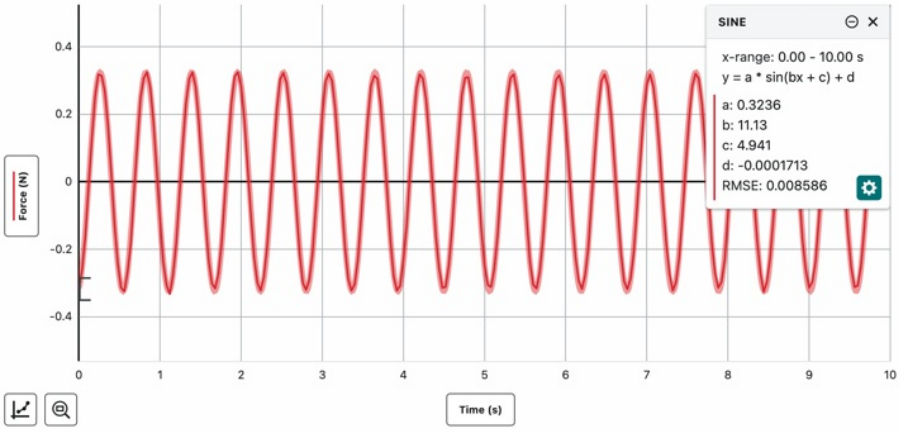
5. Inspect Force versus Time. Exclude the hand-contact interval and any sideways motion. Use a section containing several complete cycles with only modest amplitude decay for an ordinary sine fit. A visibly decaying long record should not be forced into an exact constant-amplitude sinusoid.

Before fitting: force must oscillate in time, timestamps must advance, and the amplitude must be clearly above noise. A flat line, clipping, sideways swinging or only one cycle is not an acceptable calibration record.

Fit the spring record and read the right parameter



Select a clean time interval on the force plot, open the graph tools, then choose Apply Curve Fit.



Course screenshot: a Sine fit. Read the equation shown by your software and record its parameter values and units. The screenshot is an example, not your measurement.

1. Select the clean force-time interval. Open graph tools (bottom left in the supplied screenshots), choose **Apply Curve Fit** and select **Sine / Sinusoid**. Verify that the fitted curve follows the measured cycles. Save a screenshot with the fit equation, coefficients and selected time range visible.

$$F(t) = A_F \sin(bt + \phi) + F_0$$

Define the symbols: F is measured force [N]; t is time [s]; A_F is force amplitude [N]; b is fitted angular frequency [rad/s]; φ is phase [rad]; F_0 is mean force [N]. If the app calls phase *c*, it is a fit phase, **not** the cart damping coefficient c [N·s/m].

Read the displayed equation first. In sin(bt + φ), b is rad/s and f = |b|/(2π) is Hz. In sin(2πft + φ), the fitted f is already Hz. Never multiply by 2π twice. Confirm with a manual period measurement from the graph.

Convert fitted frequency to stiffness

Why the force oscillation gives the spring frequency

Let q [m] be downward displacement from the hanging equilibrium, and Δ_0 [m] the static spring extension. At rest $k\Delta_0 = m_h g$. During a small motion, Newton's law gives:

$$m_h\ddot{q} = m_h g - k(\Delta_0 + q) = -kq$$

Gravity cancels when displacement is measured from equilibrium. In the light-damping, negligible-spring-mass approximation, the natural angular frequency is $\sqrt{(k/m_h)}$. The sensor force varies with spring extension, so it has the same oscillation frequency as q, even though its amplitude and offset differ.

$$k \approx m_h b^2, \qquad f = \frac{|b|}{2\pi}, \qquad T = \frac{2\pi}{|b|}$$

Worked example - illustrative values only

Suppose the **measured total hanging mass** is 0.200 kg and the sine fit gives b = 7.30 rad/s. Then f = 1.16 Hz, T = 0.861 s, and $k = 0.200 \times 7.30^2 =$ **10.66 N/m**. At 25 samples/s this gives about 21.5 samples/cycle. These values are a unit check, not values to copy into your report.

Independent check and acceptance

1. Read two same-sign peaks separated by N complete cycles: $T = (t_{last} - t_{first})/N$. Compare $2\pi/T$ with |b|. If they disagree more than your timing/fit uncertainty, check the fit equation, interval and units before using k.

2. Repeat with a second safe hanging mass if time permits. Stiffness should be reasonably independent of mass within uncertainty. Do not replace measured values with the nominal 10.7 N/m from the old notes.

3. If decay is substantial, the sine fit estimates the damped frequency ω_d. A measured exponential decay rate α [s^{-1}] gives $k = m_h(\omega_d^2 + \alpha^2)$ in a viscous model. If spring mass is appreciable, a uniform spring with one end fixed contributes approximately $m_s/3$ to effective moving mass; discuss this correction with the GTA rather than applying it silently.

Optional static cross-check - not a replacement for the course method

Measure spring extension Δ from a consistent unloaded reference for several safe known loads. Fit applied force $m_h g$ against Δ with an intercept; the slope is k [N/m]. Use multiple loads, not just one point. Record load, length, reference and units. This check does not require interpreting the force sensor mean as extension.

Save four spring results before assembling the cart

1. Export each force record: file/document menu at top left → **Export** → **CSV** (some versions show File → Export Data → CSV). Save one file per spring and hanging mass. Open it in MATLAB or Excel and confirm numeric Time [s] and Force [N] columns with the expected cycles.

2. Save the sine-fit screenshot as well. The CSV contains samples but may not preserve fit settings or coefficient labels. Write the fit equation and units in the notebook.

3. Only after verifying the export, clear that run or start a new experiment. Repeat the complete sequence for springs B, C and D. Check unloaded zero again if the sensor is remounted.

Spring	m_h [kg]	b [rad/s]	k [N/m]	CSV / fit image
A				
B				
C				
D				

Also record the selected fit interval, manual period check, repeat measurements and any correction or uncertainty. Label the physical springs to match this table. For independent uncertainties, the approximate propagation is:

$$\frac{u_k}{k} \approx \sqrt{\left(\frac{u_m}{m_h}\right)^2 + \left(2\frac{u_b}{|b|}\right)^2}$$

Here u_m and u_b are uncertainties in hanging mass and fitted angular frequency, and u_k is the propagated stiffness uncertainty. Fit uncertainty alone does not capture a wrong mass, sideways swinging or spring-mass effects.

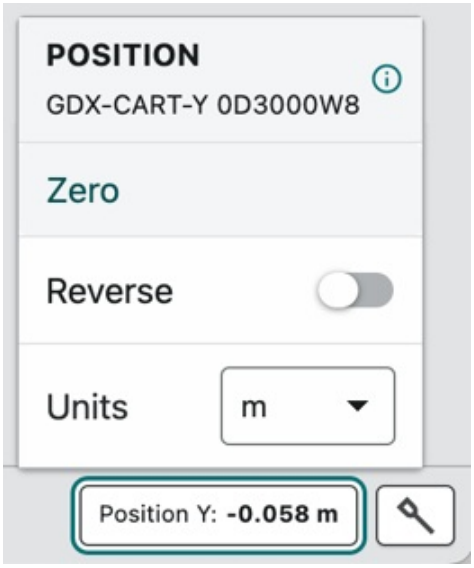
Choose the cart spring pairs

With one taut spring on each side, a cart displacement increases one spring extension and decreases the other. Both incremental forces restore the cart toward equilibrium. Consequently:

$$k_{\text{eq}} = k_L + k_R$$

k_L and k_R are the measured left/right stiffnesses [N/m]. Use the actual values for the installed IDs. Do not divide by two: this arrangement is not two springs in series. Keep both taut over the full motion. If four measured springs are nearly identical, seek the GTA-approved alternative combinations; do not invent large stiffness differences.

Zero and verify the cart position channel



Click the Position readout, confirm units m, then choose Zero while the cart is resting at equilibrium.

1. Place the cart on the level track and install the selected springs. Adjust the anchors so both are gently pre-tensioned, approximately 15 cm long near equilibrium as in the course setup. Allow the cart to settle at its natural equilibrium; do not hold it at an arbitrary track mark.

2. Power the cart. In Graphical Analysis connect by its **cart serial number** and display Position [m] versus Time [s]. Disconnect or hide the force channel if it could cause confusion.

3. With the cart stationary at equilibrium, open the **Position** readout/sensor menu and choose **Zero**. Confirm the reading is near 0 m. **Do not zero while holding the cart at the displaced release position.**

4. As a distance verification, roll the cart a measured distance along the track (for example 5 cm if safe with the installed springs). The position change magnitude should be about 0.050 m. Return to the same physical reference and check the reading returns near zero. Compare within ruler/position uncertainty; this is a verification, not an encoder scale adjustment.

5. If the sign is opposite your chosen positive direction, note the sign convention or use Reverse consistently. Do not lift and reposition the cart after zeroing: its encoder counts wheel rotation and cannot detect motion while lifted. If lifted, placed off-track or allowed to slip, return it to equilibrium and zero again.

The cart position scale is not user-calibratable. Vernier specifies zeroing, not scale calibration. A failed distance check means inspect wheel contact, rolling/slip, units, connection and track, then consult the GTA. Re-zero after changing a configuration that moves equilibrium, not after every deliberate release displacement.

Record seven controlled free-response cases

Case	Configuration	Hold fixed / record
1	Base mass; base spring pair; no brake magnets	Reference; residual friction is still present.
2-3	Two successively larger moving masses	Same spring pair; no magnets; measure total mass.
4-5	Two and four brake magnets (course configuration)	Same spring pair; include all moving brake hardware in mass.
6-7	Two other approved spring-pair combinations	Return to base mass and no added magnets. Record IDs and sum k_L + k_R.

1. For each configuration record total moving mass m [kg], spring IDs and calibrated stiffnesses, actual magnet count, and which brake parts move. Count magnets explicitly; an assembly count may differ from magnet count.

2. Check the cart zero at rest. Set an actual sampling rate no greater than 25 Hz per the supplied local procedure. Predict $f_n = \sqrt{(k_{eq}/m)}/(2\pi)$ and check samples/cycle. Start with 10 s, then extend if necessary to capture several clear cycles and decay. Record the chosen rate and duration.

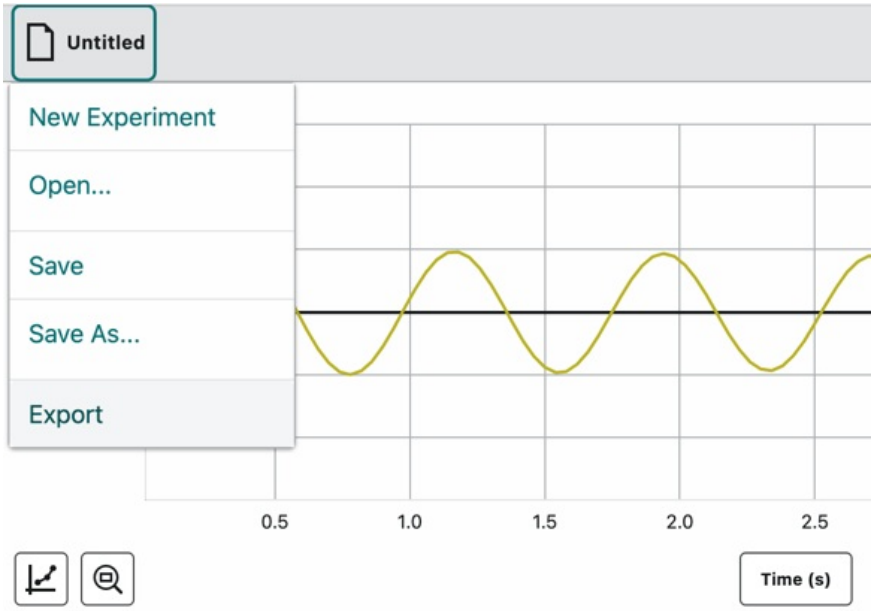
3. Move the cart a measured small distance x_0 from equilibrium. Use the same safe x_0 for comparisons unless amplitude is the intended variable. Verify both springs stay taut and cart travel will not reach end stops. **Do not re-zero at x_0.**

4. Click **Collect**, then release gently without pushing. Keep hands away. Stop after an adequate decay record or before a collision. The free-response analysis interval begins after hand contact ends.

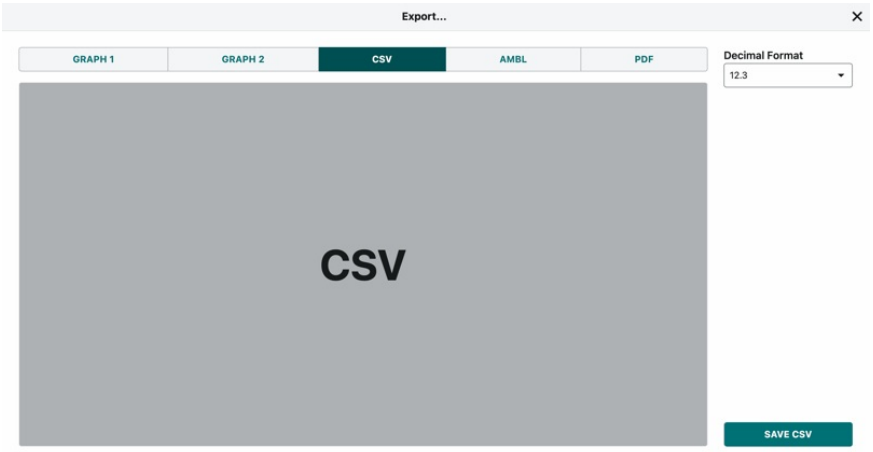
5. Save a distinct CSV immediately. Verify its time/position columns outside Graphical Analysis before clearing the run. Repeat the base run if a release, slip or connection fault is suspected.

Before moving on: open the exported file and see the expected oscillation, with time in seconds and position in stated units. Each raw file must match exactly one metadata row. Save CSVs for spring tests separately from cart tests.

Export, name and verify every run



Open the document/file menu and select Export. Saving an app project alone is not a CSV export.



Select CSV and Save CSV. Confirm the destination, file name and dataset before clearing the record.

A simple folder structure

Group##_Lab4 / Springs / CartRuns / Analysis
Examples: SpringA_m0p200kg_01.csv and Run04_m0p620kg_SA_SB_Mag2.csv. Save the fit image alongside each spring CSV. Preserve unmodified raw CSVs and keep a second backup.

Run/file	m [kg]	L/R IDs	k_eq [N/m]	magnets	x_0 [m]	f_s [Hz] / duration [s]

Extend this table to all seven cases and repeats. If the app exports more than one dataset/position column, export a single intended run or clearly identify the correct columns before analysis. A file containing force data is not a cart position record.

Turn the cart CSV into physical results

Model and definitions

After release, x [m] is cart displacement from equilibrium and t [s] is time. Total moving mass m [kg], equivalent stiffness k_{eq} [N/m] and equivalent viscous damping c [N·s/m] give:

$$m\ddot{x} + c\dot{x} + k_{\mathrm{eq}}x = 0, \qquad \omega_n = \sqrt{k_{\mathrm{eq}}/m}$$

ω_n [rad/s] is undamped natural angular frequency. For underdamped motion the displacement is an oscillation multiplied by a decaying envelope:

$$x(t) = e^{-\alpha t}[C\cos(\omega_d t) + D\sin(\omega_d t)]$$

α [s^{-1}] is decay rate; ω_d [rad/s] is damped angular frequency; C and D [m] encode initial displacement and velocity. Same-sign peak spacing gives damped period T_d [s] and frequency f_d [Hz].

$$T_d = \frac{t_{\mathrm{last}} - t_{\mathrm{first}}}{N}, \quad \omega_d = \frac{2\pi}{T_d}, \quad f_d = \frac{1}{T_d}$$

N is the number of **complete intervals** between the first and last selected same-sign peaks: five consecutive positive peaks contain four periods. Using alternating positive/negative peaks without accounting for half-periods doubles the frequency.

Let p_i be positive peak displacement [m] above equilibrium. Fit $\ln(p_i/(1\ \mathrm{m}))$ against peak time t_i; its slope is $-\alpha$. Then obtain measured undamped frequency, damping ratio ζ [dimensionless] and damping c:

$$\omega_n = \sqrt{\omega_d^2 + \alpha^2}, \qquad \zeta = \frac{\alpha}{\omega_n}, \qquad c = 2m\alpha$$

The predicted undamped frequency is $f_{n,\mathrm{theory}} = \sqrt{(k_{eq}/m)}/(2\pi)$. Compare it with measured $f_n = \omega_n/(2\pi)$. The FFT peak checks f_d; it does not directly measure damping. All time-domain amplitudes and displacement spectra remain in meters; do not apply a degree conversion.

Model limitation: the zero-magnet case still has losses. Dry/rolling friction may produce a non-exponential decay. A poor straight-line log-peak fit is evidence to discuss, not a reason to hide late points or force agreement. Strongly damped runs without enough oscillations cannot use the same peak-spacing method.

Run MATLAB and prepare the report evidence

1. Download **MIE402_Lab4_Analyze_Free_Response.m** from the Lab 4 page. Save it in a writable analysis folder. Open MATLAB, open the script and press Run. Do not run an internal helper function separately.

2. Select one **cart** CSV. Enter measured total moving mass in kg, left and right spring stiffnesses in N/m, and magnet count when prompted. The script adds the two stiffnesses; do not enter k_{eq} twice.

3. Inspect the raw plot. Enter start/end times covering only free motion after release, excluding hand contact, impacts and late noise. Confirm exported position units: choose 1 for m, 2 for cm or 3 for mm. Preserve the original CSV.

4. The script uses the final portion of the selected interval to estimate equilibrium. Ensure this portion is genuinely settled; if not, collect a longer record or consult the GTA about a justified equilibrium estimate. It subtracts offset, not an automatic linear drift.

5. Inspect selected positive peaks, log-peak regression, response fit/residuals and FFT. The current script requires at least three reliable positive peaks. Reject invalid peak choices or a flat/noisy record; do not report numerical output just because the script ran.

6. Find the saved figures and summary CSV beside the selected raw file. Verify the file stem/run identity. Repeat for every accepted configuration. Show one fully analyzed run to the GTA **before leaving**.

Evidence for the combined Labs 4-5 report

Include	Make it interpretable
Spring characterization	Four masses/fits/stiffnesses; force-time fit image; units, interval and uncertainty/check.
Cart time trace	Displacement, same-sign peaks, fitted envelope and run metadata.
Decay/model evidence	Log-peak fit with α and fit quality; fitted response and residuals.
Frequency spectrum	Frequency axis [Hz], displacement spectrum units and annotated peak value.
Seven-case summary	m, spring IDs, k_eq, magnets, f_d, FFT peak, measured/predicted f_n, ζ, α and c with units.
Discussion	Mass/stiffness/damping trends; uncertainty, nonideal friction and reasons for mismatch.

Use only equations needed for the calculations and interpretation. Define every symbol the first time it appears. An agreement metric may be $100 \times |f_{n,measured} - f_{n,theory}| / f_{n,theory}$; state this definition. Damping hardware can also change moving mass, so explain both effects.

Troubleshooting and exit checklist

Symptom	Check and action
Force reads zero / flat line	Confirm force sensor and Force channel, not cart Position. Apply/remove a safe load. Check connection and ask GTA if the sensor remains unresponsive.
Spring k is implausible	Check kg versus g, b in rad/s versus f in Hz, hanger mass, fit interval and selected spring. Do not substitute the nominal value.
Sine fit misses the cycles	Remove hand contact, include several cycles, verify vertical motion and fit model. Use a shorter lightly decaying interval; save the rejected evidence if useful.
Cart position wrong after lifting	Wheel encoder cannot detect lifted motion. Replace on track, settle at equilibrium and zero again.
Cart x(t) starts near zero then shifts	Was zero set at the displaced release point? Zero at equilibrium instead; preserve/repeat the run as appropriate.
Few positive peaks / fit fails	Check units, interval, equilibrium, noise and strong damping. Do not invent peaks; repeat or ask GTA for another analysis approach.
Frequency or decay changes unexpectedly	Check actual total moving mass, spring IDs, tautness, magnets, wheel slip, track level and a pushed release.
Export will not analyze	Open CSV; identify Time and Position columns and units, numeric timestamps and one intended run. Preserve raw data.

Before you leave

Confirm that you have: **(1)** four labeled spring stiffness results with masses and fit evidence; **(2)** seven accepted cart CSVs with complete metadata; **(3)** verified exports and a second backup; **(4)** one complete MATLAB analysis shown to the GTA; and **(5)** recorded anomalies and uncertainty sources. Return carts and sensors to the charging stations after finishing.

Source notes and device manuals

Procedure and supplied screenshots: MIE 402 LAB4_MIE402.pdf (prior course handout, pp. 1-3) and Lecture Notes/13_Lab4_Exp.pdf. The acquisition sequence, four springs, local sampling limit and seven-case requirement are retained. New diagrams and worked numbers are instructional examples, not measured results.

Vernier Go Direct Sensor Cart manual: position encoder, zeroing and force calibration information. Go Direct Force and Acceleration manual: stored force calibration and model-specific checks. Confirm the actual force-sensor model before any calibration adjustment.

Course Lab 4 page and MATLAB downloads. Interface screenshots show the supplied app version; labels may move in a later version. Use the named control, correct channel and physical check rather than a fixed screen coordinate.